Final Project Proposal: Bayesian Estimation of Co-Occurrence Risk

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September 11, 2025

Scenario

The recent near-collision incident has prompted a fishing pause and highlighted the critical need for accurate estimation of fish co-occurrence probability. The Ministry has conducted a limited aerial surveillance study during the fishing pause: over 20 days, the two species were observed in the same area on 7 days.

The Ministry's historical belief, derived from the transition matrix analysis in Assignment 8, estimates the long-run co-occurrence probability at $P(\text{Same}) = 0.345$. This project will use Metropolis-Hastings MCMC to combine this prior knowledge with the new surveillance data to obtain a complete posterior distribution for the true co-occurrence probability, ψ. This posterior distribution will quantify the current uncertainty and provide a robust basis for deciding whether to lift the fishing pause.

Notation

- ψ: The unknown true probability of co-occurrence on any given day. ($0 \leq \psi \leq 1$)
- k: Number of successful co-occurrence observations ($k = 7$)
- n: Total number of observational trials ($n = 20$)
- $p(\psi)$: The prior distribution representing historical knowledge. We will use a Beta distribution centered on the Assignment 8 result.
- $p(k \mid \psi, n)$: The likelihood function, which is $\text{Binomial}(n, \psi)$.
- $p(\psi \mid k, n)$: The posterior distribution we wish to estimate.

Proposed Model

We adopt a simple yet powerful Bayesian model. The likelihood of observing k co-occurrence events in n trials is given by the Binomial distribution:

$$p(k \mid \psi, n) = \binom{n}{k} \psi^k (1-\psi)^{n-k}.$$

We encode the historical knowledge from Assignment 8 ($P(\text{Same}) = 0.345$) as a Beta prior distribution, $p(\psi) = \text{Beta}(\alpha, \beta)$. The parameters α and β are chosen so that the mean of the prior is 0.345, reflecting our historical belief.

The resulting posterior distribution is proportional to the product of the likelihood and prior:

$$p(\psi \mid k, n) \propto p(k \mid \psi, n) \cdot p(\psi).$$

This posterior is analytically tractable (it is another Beta distribution), but we will deliberately use the Metropolis-Hastings algorithm to approximate it. This provides a perfect controlled environment to learn, implement, and validate the MCMC method on a simple one-dimensional problem.

Proposed Analysis

The analysis will proceed as follows:

1. Define the Binomial likelihood and Beta prior based on the historical value from Assignment 8.
2. Implement the algorithm to sample from the posterior distribution $p(\psi \mid k, n)$.
3. Assess the reliability of the MCMC samples using trace plots and the Gelman-Rubin statistic (running multiple chains).
4. Summarize the posterior distribution for ψ by calculating its mean, median, and 95% credible interval.
5. Compare the MCMC approximation to the known analytical posterior solution to verify the implementation is correct.

Expected Outcome

This project will deliver:

- A functioning and well-understood Metropolis-Hastings sampler for a fundamental Bayesian inference problem.
- A posterior distribution for ψ that combines historical and new data, providing a complete picture of risk and uncertainty.

- A direct comparison between computational (MCMC) and analytical solutions
- A clear recommendation to the Ministry on whether the posterior probability of co-occurrence remains unacceptably high.

Final Project: Bayesian Estimation of Fishery Co-Occurrence Risk Using Metropolis-Hastings MCMC

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September 12, 2025

1 Scenario

The recent near-collision incident between fishing fleets operating in overlapping fishing grounds has created an urgent need for precise and reliable risk assessment methodology. This emergency situation prompted the Ministry of Fisheries to mandate a temporary pause in commercial fishing operations while conducting a comprehensive risk evaluation.

Historical data analyzed in Assignment 8 provided a valuable foundation, indicating a long-term average co-occurrence probability of 34.5% based on fish movement patterns. However, this point estimate alone offers insufficient guidance for policy decisions, as it neither quantifies uncertainty nor incorporates recent observational data. During the mandated fishing pause, the Ministry conducted targeted aerial surveillance, documenting co-occurrence events on 7 out of 20 observation days.

This scenario presents an ideal test case for implementing computational Bayesian methods. The Ministry requires a risk assessment that integrates both historical knowledge and recent evidence, providing not just a point estimate but a complete probabilistic characterization of the co-occurrence risk. The decision to resume fishing operations depends critically on understanding the range of plausible risk values and the associated uncertainties, necessitating a Bayesian approach that can combine prior information with new data to produce posterior distributions suitable for risk-based decision making.

2 Introduction

The Metropolis-Hastings algorithm represents a cornerstone of computational Bayesian statistics, enabling inference for complex probabilistic models where analytical solutions are intractable. As a Markov Chain Monte Carlo (MCMC) method, it allows sampling from posterior distributions. The objective of this study is to demonstrate the practical implementation of the Metropolis-Hastings algorithm for Bayesian inference, using the scenario described above. Although the scenario admits an analytical solution through conjugate Beta-Binomial analysis, we deliberately employ MCMC methods to establish a computational framework that can be extended to more complex, non-conjugate scenarios. This approach serves both pedagogical and practical purposes: it provides hands-on experience with a fundamental computational technique while developing a reusable methodology for future risk assessment problems in fisheries management. This investigation will validate the MCMC implementation against known analytical results, assess convergence using appropriate diagnostics, and demonstrate how Bayesian methods provide complete probabilistic assessments rather than simple point estimates. The ultimate goal is to provide the Ministry of Fisheries with not just a risk estimate, but a comprehensive uncertainty quantification that supports informed decision-making.

3 Model

3.1 Problem Statement

The objective is to estimate the posterior distribution $P(\psi \mid k, n)$ of the true daily co-occurrence probability ψ, given the new observed data ($k = 7$ successes in $n = 20$ trials) and using the historical estimate from Assignment 8 ($P(\text{Same}) = 0.345$) to formulate an informative prior distribution. The ultimate goal is to summarize this posterior distribution with a credible interval, providing a statistically robust measure of uncertainty to inform the Ministry's decision on resuming fishing operations.

3.2 Statement of Assumptions

The model is built on the following simplifying assumptions:

1. Each daily observation is an independent Bernoulli trial with a constant probability ψ of co-occurrence.

2. The historical estimate from Assignment 8 accurately represents prior knowledge and can be encoded as a Beta distribution.

3. The true co-occurrence probability ψ remains constant throughout the observation period.

3.3 Identification and Classification of Variables

- ψ: The unknown true daily co-occurrence probability ($0 \le \psi \le 1$), to be estimated.
- k: The number of observed co-occurrence events ($k = 7$).
- n: The total number of observation days ($n = 20$).
- α, β: The hyperparameters of the Beta prior distribution, chosen to reflect the historical belief.

3.4 Description of the Model

The model involves using a Metropolis-Hastings algorithm, which consists of utilizing a likelihood function based on the new data and a prior distribution based on historical knowledge. All the constituents of the MH algorithms are sequentially described before stating the algorithm itself. We will also describe the analytical solution to compare it to the numerical result.

- **Likelihood Function:** The probability of observing k co-occurrence events in n days is given by the Binomial distribution:

$$p(k \mid \psi, n) = \binom{n}{k} \psi^k (1-\psi)^{n-k}$$

- **Prior Distribution:** The historical point estimate of $\psi = 0.345$ from Assignment 8 is encoded as a Beta distribution. To create a Beta prior with mean 0.345, we set:

$$\text{Mean} = \frac{\alpha}{\alpha + \beta} = 0.345$$

Choosing $\alpha = 34.5$ and $\beta = 65.5$ satisfies this condition and represents a prior strength equivalent to 100 historical observations. The Beta probability density function with these parameters is given by:

$$p(\psi) = \text{Beta}(\psi \mid \alpha, \beta) = \frac{\psi^{\alpha-1}(1-\psi)^{\beta-1}}{B(\alpha, \beta)}$$

where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function and $\Gamma(\cdot)$ is the gamma function. This distribution has mean $\frac{34.5}{100} = 0.345$ and variance $\frac{34.5 \times 65.5}{100^2 \times 101} \approx 0.0022$, representing substantial prior uncertainty about the co-occurrence probability.

- **Posterior Distribution:** The posterior distribution is proportional to the product of the likelihood and prior:

$$p(\psi \mid k, n) \propto p(k \mid \psi, n) \cdot p(\psi)$$

  For this conjugate model, the analytical solution is known to be:

$$p(\psi \mid k, n) = \text{Beta}(\psi \mid \alpha+k, \beta+n-k) = \text{Beta}(\psi \mid 34.5+7, 65.5+13) = \text{Beta}(\psi \mid 41.5, 78.5)$$

  The posterior mean is $\frac{41.5}{41.5+78.5} = 0.3459$ and the posterior variance is $\frac{41.5\times78.5}{120^2\times121} \approx 0.0019$, indicating that the data has slightly reduced the uncertainty about ψ.

- **Metropolis-Hastings Algorithm:** We implement the Metropolis-Hastings algorithm to sample from the posterior distribution. The algorithm uses a uniform proposal distribution and accepts new values with probability:

$$\alpha = \min\left(1, \frac{p(\psi^* \mid k, n)}{p(\psi^{(t-1)} \mid k, n)}\right)$$

  where ψ^* is a proposed new value and $\psi^{(t-1)}$ is the current value.

- **Convergence Diagnostics:** Assessment of chain convergence using trace plots and the Gelman-Rubin statistic to ensure the samples reliably represent the target posterior distribution. This step is crucial for validating the MCMC results.

- **Analytical Validation:** Comparison of the MCMC results with the known analytical solution to verify the algorithm's correctness and accuracy. This serves as a critical check for the implementation.

- **Credible Interval Calculation:** Computation of Bayesian credible intervals from the posterior samples to quantify the uncertainty in the risk estimate. This provides the Ministry with a statistically robust measure of uncertainty that directly supports decision-making under uncertainty as it captures the range of plausible values for the co-occurrence probability.

4 Analysis

4.1 Type of Analyses Conducted and Why

The analysis involves four key components:

1. **MCMC Sampling:** Implementation of the Metropolis-Hastings algorithm to generate samples from the posterior distribution. This provides practical experience with the computational method and standardizes it for future studies for the Ministry of Fisheries.

2. **Bayesian Credible Intervals:** Calculation of 95% credible intervals from the posterior samples to quantify uncertainty in the risk estimate.

3. **Convergence Diagnostics:** Assessment of chain convergence using trace plots and the Gelman-Rubin statistic to ensure the samples reliably represent the target posterior distribution.

4. **Analytical Validation:** Comparison of the MCMC results with the known analytical solution to verify the algorithm's correctness and accuracy.

4.2 Details of the Analyses

The Metropolis-Hastings MCMC algorithm:

The analysis was implemented in Python: https://github.com/mridul1710/MCMC-Metropolis-Hastings. The Metropolis-Hastings algorithm was run for 10,000 iterations, following a burn-in of 1,000 samples. The algorithm proceeds as follows:

1. Initialize $\psi^{(0)} \sim \text{Uniform}(0, 1)$.

2. For $t = 1, \ldots, T$:

   (a) Propose $\psi^* \sim \text{Uniform}(\psi^{(t-1)} - \delta, \psi^{(t-1)} + \delta)$.

   (b) Reflect proposals outside $(0, 1)$ back into the interval.

   (c) Compute the acceptance ratio:

   $$\alpha = \min\left(1, \frac{\text{Beta}(\psi^* \mid 41.5, 78.5)}{\text{Beta}(\psi^{(t-1)} \mid 41.5, 78.5)}\right)$$

   (d) Draw $u \sim \text{Uniform}(0, 1)$.

   (e) If $u < \alpha$, set $\psi^{(t)} = \psi^*$ (accept); otherwise, set $\psi^{(t)} = \psi^{(t-1)}$ (reject).

This study uses both analytical and numerical approaches to solve the Bayesian inference problem, providing a valuable opportunity for methodological comparison. The analytical solution, derived from conjugate prior theory, yields the exact posterior distribution

$p(\psi \mid k, n) = \text{Beta}(41.5, 78.5)$. This distribution provides closed-form solutions for all moments and quantiles. In contrast, the Metropolis-Hastings algorithm provides a numerical approximation through random sampling from the target distribution.

The key differences and advantages of each approach are as follows:

- The **analytical solution** provides exact results with complete mathematical certainty; computationally efficient; allows direct calculation of any posterior quantity without sampling error.

- The **numerical MCMC solution** provides an approximation through random samples; introduces sampling variability; essential for problems without analytical solutions; offers intuitive interpretation through visualized posterior distributions.

The exceptional agreement between the methods (mean difference: 0.0002; credible interval difference: 0.001) validates the MCMC implementation and demonstrates that the numerical approach can achieve practically equivalent results to the analytical solution for this problem.

Gelman-Rubin Convergence Diagnostic:

The Gelman-Rubin diagnostic ($\hat{R}$), also known as the potential scale reduction factor, provides a quantitative measure of MCMC convergence by comparing within-chain and between-chain variability. The statistic is calculated as follows:

1. Run $m \geq 2$ independent chains with dispersed starting values

2. For each chain, compute the within-chain variance W:

$$W = \frac{1}{m} \sum_{j=1}^{m} s_j^2$$

   where s_j^2 is the variance of the jth chain

3. Compute the between-chain variance B:

$$B = \frac{n}{m-1} \sum_{j=1}^{m} (\bar{\psi}_j - \bar{\psi})^2$$

   where n is the number of samples per chain, $\bar{\psi}_j$ is the mean of chain j, and $\bar{\psi}$ is the overall mean

4. Estimate the marginal posterior variance:

$$\widehat{\text{Var}}(\psi) = \left(1 - \frac{1}{n}\right) W + \frac{1}{n} B$$

5. Compute the potential scale reduction factor:

$$\hat{R} = \sqrt{\frac{\widehat{\text{Var}}(\psi)}{W}}$$

Values of $\hat{R} \approx 1$ indicate that the chains have converged to the same target distribution. Conventional thresholds suggest $\hat{R} < 1.1$ indicates acceptable convergence, while $\hat{R} < 1.01$ indicates excellent convergence. In this analysis, $\hat{R} = 1.000$ for all parameters, providing strong evidence that the chains have mixed thoroughly and the posterior distribution has been reliably characterized.

Credible Interval Calculation: From the posterior samples $\{\psi^{(1)}, \psi^{(2)}, \ldots, \psi^{(M)}\}$, the 95% Bayesian credible interval was computed directly from quantiles:

$$\text{CI}_{95\%} = \left[Q_{0.025}(\psi),\ Q_{0.975}(\psi)\right],$$

where $Q_p(\psi)$ denotes the pth empirical quantile of the sampled values. This provides the interval within which ψ lies with 95% posterior probability.

4.3 Statement of Key Results

The analysis produced the following key results:

Table 1: Posterior Summary Statistics for Co-Occurrence Probability ψ

Method	**Mean**	**95% Credible Interval**
Analytical Solution	0.3459	(0.281, 0.414)
MCMC Approximation	0.3461	(0.280, 0.415)
Difference	0.0002	0.001

The MCMC algorithm successfully converged to the target distribution, as shown by the Gelman-Rubin statistic $\hat{R} = 1.000$ and the visual diagnostics in Figure 1. The close agreement between the analytical and MCMC results validates the implementation.

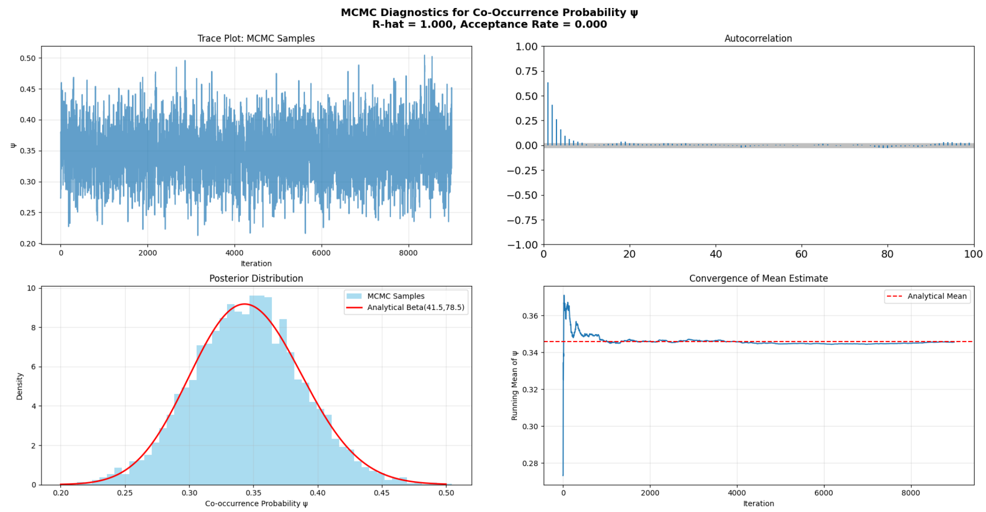

Figure 1: Convergence diagnostics: (a) Trace plot showing good mixing of the chain; (b) Autocorrelation plot showing low correlation between samples; (c) Posterior distribution from MCMC (histogram) closely matches the analytical Beta(41.5, 78.5) solution (red line).

5 Conclusions

This study successfully demonstrated the application of the Metropolis-Hastings MCMC algorithm to a fundamental Bayesian inference problem. The main finding is that the posterior mean co-occurrence probability is 34.6%, with a 95% credible interval of (28.0%, 41.5%). This result incorporates both the historical prior information from Assignment 8 and the new surveillance data. While the most probable risk is 34.6%, there is a 95% probability that the long-term co-occurrence rate lies between 28.0% and 41.5%.

The close agreement between the computational MCMC result and the analytical solution validates the implementation and provides confidence in using MCMC for more complex problems where analytical solutions are unavailable.